



**Department of Electrical, Computer
& Biomedical Engineering**
Faculty of Engineering & Architectural Science

Course Title:	Fundamentals of Data Engineering
Course Number:	COE848
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Instructor:	Dr. Yasaman Ahmadiadli
Lab Report Number	Lab 1
Lab Title	Lab 1: Setup and Project Specification

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The objective of this project is to build a Restaurant Management system that maintains a database of different restaurants and their respective table availability, reservations, ratings, food order menu, opening and closing hours, and other factors. The primary intention of building this management system is to keep a neat and organized track of the many restaurants around Toronto area(GTA), helping restaurant owners and managers to track orders, manage busyness and to book reservations very efficiently, allowing the restaurant owners to better understand their own restaurants and have insights to make meaningful operational decisions. Not only that, but this management system is just as useful and valuable to consumers too; customers, locals, and people that just want to explore Toronto can have an easier access to the restaurants around them as well as accurate insights about the restaurant and of course a better idea of what kind of restaurants are near them.

The database will store information and details based on different restaurants; it will specifically include attributes like their name, location, hours, cuisine type, and seating capacity. Restaurants will also have the option to be more specific and set up different types of tables, each table having a number associated with it, seating capacity of the table itself, and the availability status. This gives the restaurant owners the ability to control and manage crowds and busyness while making sure that the restaurant is not overflowing with customers. It also gives them an idea of what hours and days people come most often, giving them statistics and information that can assist them in making great business decisions.

The database will also store customer information, such as customer name, phone number, reservation details or even reservation history. The order itself is also an entity and will be associated with a specific reservation number, also having date of order, and total amount to be paid. Each order can also be composed of “Order Items”, which associate with the “Menu Items”, in this case we can make them the same thing such that every item has a price, name, rating, availability and even quantity to it. This again allows restaurant owners to understand which items from their restaurant are demandable and great.

Similarly reservations will work in the same way, except they will be associated with a given restaurant and customer entity, and will include table info, number of guests, and date. The management system will allow restaurants to practically control and manage all their restaurant services accurately and efficiently.

As it could be seen, the Restaurant Management System is a very realistic and valuable database application that will reflect real-life business prospects. This project has the complexity and scalability to demonstrate the concepts and use of relational databases that we learn within our class.

A list of queries that I would like to be answered by the manage system database:

- Which of the Restaurants that are above 4 stars have availability for reservations on Tuesday?
- Which of the following restaurants have “chicken” menu items with the lowest order price
- Which of the Restaurants are available for reservations on Friday evening for a spot for 10 people

- Is the restaurant's highest rated food available for ordering?
- How many restaurants are there in a specific location for Japanese cuisine food?
- What is the highest rated Mexican restaurant nearby?
- Which hours is this specific restaurant operating within?
- Which branch of the restaurant has the highest rating, and what is its location?
- How many menu/order items does a store have that is below a certain price?
- Can I reserve this specific table for 12 people on Friday evening for this specific item(if it's available)?